



Santiago Aguilar Mello

Student at Tecnológico de Monterrey

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Profile

Internationally experienced CS student with hands-on exposure to IoT systems, C++, Python, and web development. Strong in team collaboration, multicultural communication, and technical teaching. Passionate about embedded systems and applied computing.

Technical Skills

Programming:	C++, Python, HTML/CSS, Arduino (ESP32), JavaScript (basic)
Tools:	Git/GitHub, Linux/Bash, VSCode, MQTT, ThingSpeak API, ESP-IDF, Excel
Soft Skills:	Leadership, communication, mentoring, teamwork, curriculum design

Education

Tecnológico de Monterrey (ITESM)

2024–Present

B.S. in Computer Science and Technology — Graduate of the Peer Mentorship Program. *Relevant coursework:* Data Structures, OOP (C++), IoT, Embedded Systems, Device Interconnection, Databases, Software Engineering, Web Development, Differential Equations, Computational Thinking.

Rotterdam International Secondary School (RISS), Netherlands

2022–2024

International Baccalaureate Diploma — **33/45**. Awards: Excellence in English, high academic performance, leadership. Head of Charity: led campaigns including “Buy a Tulip, Help a Girl” and Red Cross disaster relief for Turkey & Syria.

Technical Projects

DT Construct ICS — Freelance Client Website

dte-ingenieria.com

- Designed and shipped a live multi-page website for a Mexican construction company, hand-coded in HTML, CSS, and JavaScript with interactive Leaflet maps and a six-section services catalog.

IoT Smart Parking System — ESP32, Sensors, MQTT, Python

- Built a full IoT prototype with occupancy detection, automated barrier control, and cloud services.

Data Structures in C++

- Implemented BSTs, AVL, heaps, queues, hash logs, BFS/DFS, Dijkstra, including full complexity analysis.

IntercambiaTec SRS

- Developed an IEEE-style SRS with UML diagrams, use cases, and functional requirements.

Python Educational Tools

- Created programming mini-games for 12–16 year-olds; used in classroom instruction.

Work Experience

English Teacher — ProEnglish Rotterdam Summer Camp

Aug 2025–Present Taught 165 children (ages 4–12) using

immersive, play-based English learning.

Python Instructor (Ages 12–16)

Feb 2025–Jul 2025 Designed and taught a Python curriculum with interactive coding

challenges.

Volunteer & Peer Mentor — Tec de Monterrey

2024–Present Assisted new students academically and socially; guided

them through university adaptation.

Informal Babysitter

2021–2024 Supported language acquisition for children (7–9) using immersive English activities.

Extracurricular Activities

COPARMEX ITESM CSF • Peer Mentorship Program Graduate • Charity leadership (RISS)

Languages

Spanish (native), Portuguese (native), English (C2), Dutch (A1)

Interests

IoT, embedded systems, AI in education, cloud computing, theatre, public speaking, volunteering.